

Varun Srinivasan

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Professional Summary

Computer Science undergraduate with experience building AI-driven applications, cloud-native systems, and blockchain solutions. Strong foundation in data structures, machine learning, and full-stack development.

Education

Vellore Institute of Technology, Chennai, Computer Science and Engineering	2023 – 2027
CGPA: 9.32/10	
Maharishi Vidhya Mandir, Hosur	Class XII: 91% 2023

Technical Skills

- **Programming:** Python, Java, C++, C, SQL, R
- **AI/ML:** PyTorch, TensorFlow, Scikit-learn, GANs, CNNs, Quantum ML, NLP, Vision-Language Models (VLMs)
- **Web, Backend & Cloud:** Flask, React, Next.js, HTML, CSS, JavaScript, AWS, Docker
- **Blockchain:** Solidity, Ethereum, IPFS, Ethers.js
- **Databases & Tools:** Oracle, VS Code, Git, GitHub, RStudio, Pandas, NumPy, Matplotlib

Publications

- **Enhancing Crop Recommendation Accuracy using Quantum Support Vector Machines on Complex Agricultural Datasets**
Accepted at IEEE International Conference (ETAACT 2026) | DOI: 10.1109/ETAACT69135.2026.11541506
- **CNN-Based Adaptive Image-in-Image Steganography Using Learned Residual Embedding**
Accepted at IEEE International Conference (AIDML 2026)

Projects

- **Crop Recommendation** *Research Project | GitHub*
Designed a quantum-enhanced crop recommendation system integrating Classical SVM, QSVM, and noise-aware QSVM with geospatial weather/soil APIs. Containerized the Flask backend using Docker and deployed on AWS.
- **FarmFlow Enhanced** *Personal Project | GitHub*
Developed a decentralized agriculture platform integrating Ethereum smart contracts, Dynamic Crop Risk Index (DCRI) computation, IPFS-backed storage, and AI-based crop analytics. Built an off-chain Flask service for weather and disease analysis that updates on-chain insurance and pricing logic.
- **Adaptive Fusion of CNN and Swin Transformer for Metal Defect Classification** *Research Project | GitHub*
Designed a hybrid CNN-Transformer architecture combining ResNet-50 and Swin Transformer through a Multi-Head Attention Feature Fusion module for industrial surface defect classification. Applied Focal Loss and temperature scaling for confidence calibration, achieving 99.3% accuracy with significantly reduced Expected Calibration Error.

Experience

- **Intern | Summer Research Internship Programme 2025** *May 2025 – August 2025*
 - Fine-tuned the Qwen2.5 Vision-Language Model for image captioning using parameter-efficient training techniques and evaluated model performance with BLEU, METEOR, SPICE, and CIDEr metrics.
- **Member | CloudOps Club | VIT, Chennai** *2024 – 2025*
 - Member of the Content department, created marketing materials and content for club events.

Certifications

- **Machine Learning Course** — Google for Developers
- **AWS Certified Cloud Practitioner (CLF-C02)** — Amazon Web Services
- **AWS Certified Solutions Architect – Associate (SAA-C03)** — Amazon Web Services